Practical - 2

To implement the job sequencing problem using a greedy algorithm to maximize profit.

```
#include <iostream>
#include <algorithm>
using namespace std;

struct Job {
    char id;
    int deadline;
    int profit;
};
bool compare(Job a, Job b) {
    return a.profit > b.profit;
}

int main() {
    Job jobs[] = {
        {'A', 2, 100},
        {'B', 1, 50},
        {'C', 2, 10}
    };

    int n = 3;
    sort(jobs, jobs + n, compare);

    int maxDeadline = 0;
    for (int i = 0; i < n; i++) {
        maxDeadline = max(maxDeadline, jobs[i].deadline);
    }

    char slots[10];
    for (int i = 0; i < maxDeadline; i++) {
        slots[i] = '-';
```

```
    }
    int totalProfit = 0;

    for (int i = 0; i < n; i++) {
        for (int j = jobs[i].deadline - 1; j >= 0; j--) {
            if (slots[j] == '-') {
                slots[j] = jobs[i].id;
                totalProfit += jobs[i].profit;
                break;
            }
        }
    }
    cout << "Selected Jobs: ";
    for (int i = 0; i < maxDeadline; i++) {
        if (slots[i] != '-') {
            cout << slots[i] << " ";
        }
    }

    cout << "\nMaximum Profit: " << totalProfit << endl;

    return 0;
}
// OUTPUT
```

